GRAPH THEORY:
HOMEWORK 7
BOOK SECTION 3.1 PART III AND 4.1.

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1. Let M be a maximal matching and L a minimal edge cover in a graph with no isolated vertices. Probe the following statements below:

(a) M is a maximal matching if and only if $M$is contained in a minimum edge cover.
(b) L is a minimum edge cover if and only if L contains a maximal matching.

Proof. □

2. Let G be a simple graph in which the sum of degrees of any k vertices is less than $n-k$. Prove that every maximal independent set in G has more than k vertices.

Proof. □

3. An edge e of a graph G is α-critical is $\alpha(G-e) > \alpha(G)$. Suppose that xy and xz are α-critical edges in G. Prove that G has an induced subgraph that is an odd cycle containing xy and xz. (Hint: Let Y, Z be maximum independent sets in $G-xy$ and $G-xz$ respectively. Let $H = G[Y \Delta Z]$. Prove that every component of H has the same number of vertices from Y to Z. Use this to prove y and z belong to the same component of H.)

Proof. □

4. Let G be a connected graph with at least three vertices. Form G' and G by adding an edge with endpoints x, y whenever $d_G(x, y) = 2$. Prove that G' is 2-connected.

Proof. □

5. Prove that $\kappa(G) = \delta(G)$ if G is simple and $\delta(G) \geq n(G) - 2$. Prove that this is the best possible for each $n \geq 4$ by constructing a simple n-vertex graph with minimum degree $n-3$ and connectivity less than $n-3$.

Proof. □

Date: October 15, 2026.